Automating Data Import and Relationship Mapping using Import Sets & Dot Walking

ABSTRACT

Modern organizations handle large volumes of data that must be imported from external sources such as Excel or CSV files. Manual data entry is time-consuming and may lead to errors. To overcome this problem, automation techniques are used.

This project focuses on automating data import using Import Sets and establishing relationships between tables using Dot Walking. Import Sets allow external data to be uploaded into a staging table before being transferred into the target table using Transform Maps. Dot Walking helps access fields from related tables easily using reference fields.

The system ensures faster data processing, reduces manual work, and improves data accuracy. This project demonstrates how automated data import and relationship mapping can be implemented efficiently in the platform.

CHAPTER 1

INTRODUCTION

1.1 Introduction

Organizations frequently import large datasets into their systems. These datasets may come from spreadsheets, external databases, or third-party applications.

Manually entering such data is inefficient and increases the chances of errors. To solve this problem, automation techniques are used.

This project demonstrates how to automate the process of importing data using Import Sets and how to map relationships between tables using Dot Walking.

1.2 Objectives

The main objectives of this project are:

- To automate data import from external files.
- To reduce manual data entry.
- To map relationships between tables.
- To access related data using Dot Walking.

1.3 Scope

The scope of the project includes:

- Importing data from Excel files.
- Transforming data to target tables.
- Mapping relationships using reference fields.
- Accessing related data using Dot Walking.

CHAPTER 2
SYSTEM ANALYSIS

2.1 Existing System

In traditional systems, data is manually entered into the database.

Problems in the existing system:

- Time consuming
- High chance of errors
- Difficult to maintain relationships
- Manual data processing

2.2 Proposed System

The proposed system automates the process of importing data using Import Sets and Transform Maps.

Advantages:

- Faster data processing
- Accurate data import
- Reduced manual effort
- Easy relationship mapping

CHAPTER 3
SYSTEM ARCHITECTURE

Architecture Diagram

```
+-------------------+
|  Excel / CSV      |
|  Data Source      |
+---------+---------+
          |
          v
+-------------------+
|    Import Set     |
| (Staging Table)   |
+---------+---------+
          |
          v
+-------------------+
|  Transform Map    |
| Field Mapping     |
+---------+---------+
          |
          v
+-------------------+
|   Target Table    |
|  (sys_user)       |
+---------+---------+
          |
          v
```

```
+-------------------+
| Dot Walking       |
| Relationship Data |
+-------------------+
```

Explanation:

1. Data is uploaded from an Excel file.

2. The file is stored in the Import Set table.

3. Transform Map moves the data to the target table.

4. Dot Walking is used to access related records.

CHAPTER 4
SYSTEM IMPLEMENTATION

Step 1 – Data Source Upload

An Excel file containing employee data is uploaded into the system.

The system creates an Import Set table where the data is temporarily stored.

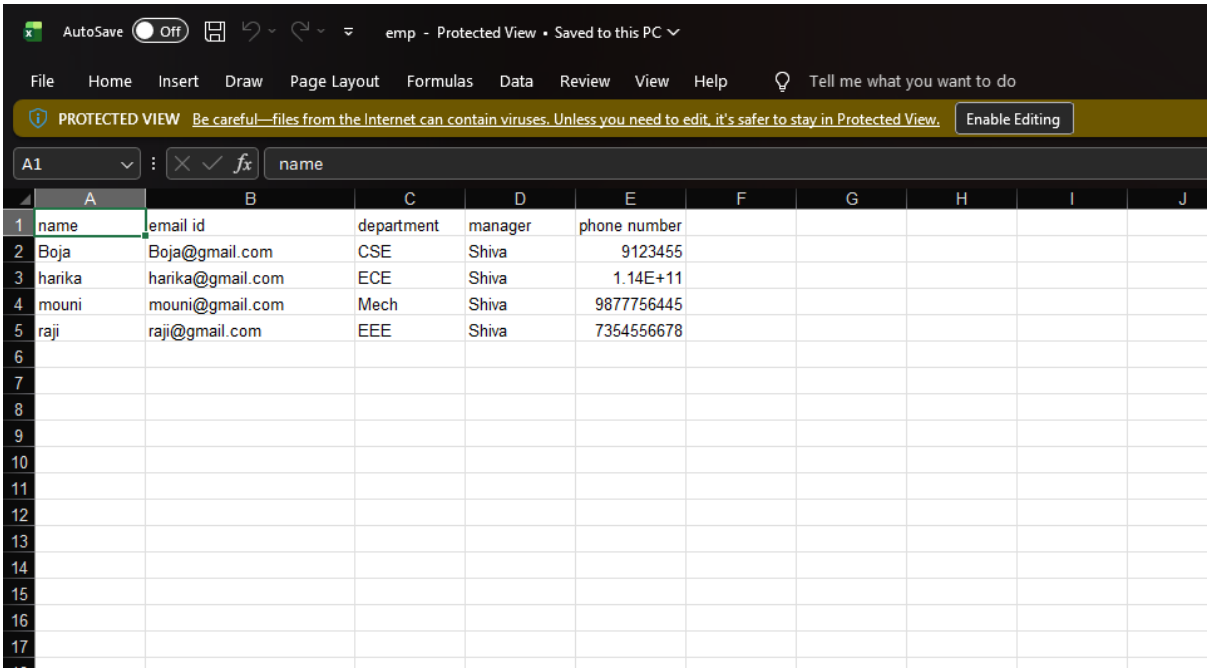
Example fields in Excel file:

Name

Email

Department

Phone

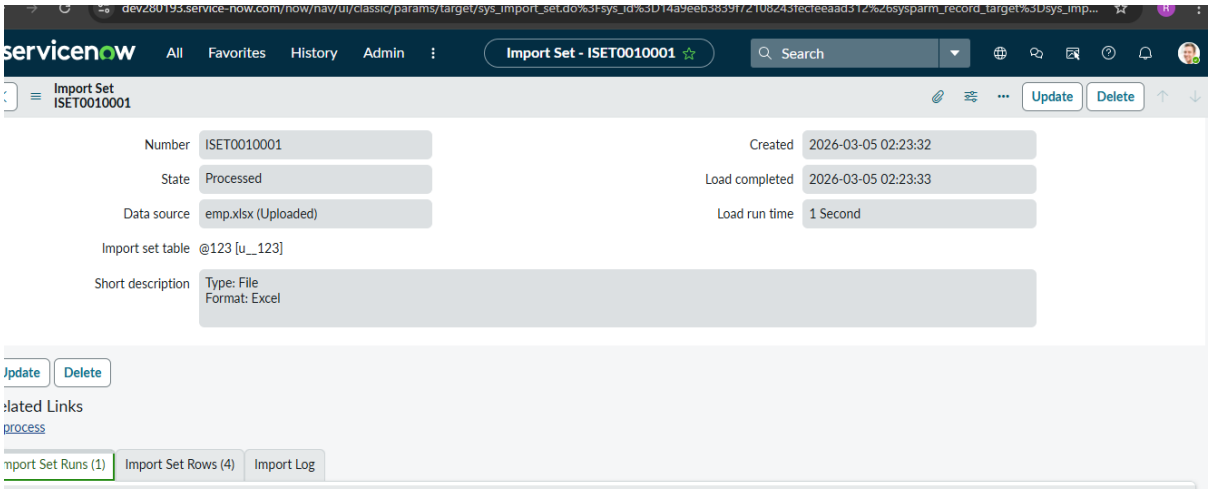


	A	B	C	D	E
1	name	email id	department	manager	phone number
2	Boja	Boja@gmail.com	CSE	Shiva	9123455
3	harika	harika@gmail.com	ECE	Shiva	1.14E+11
4	mouni	mouni@gmail.com	Mech	Shiva	9877756445
5	raji	raji@gmail.com	EEE	Shiva	7354556678

Step 2 – Import Set Creation

When the file is uploaded, the system automatically creates a staging table where the imported data is stored.

This staging table is called an Import Set Table.



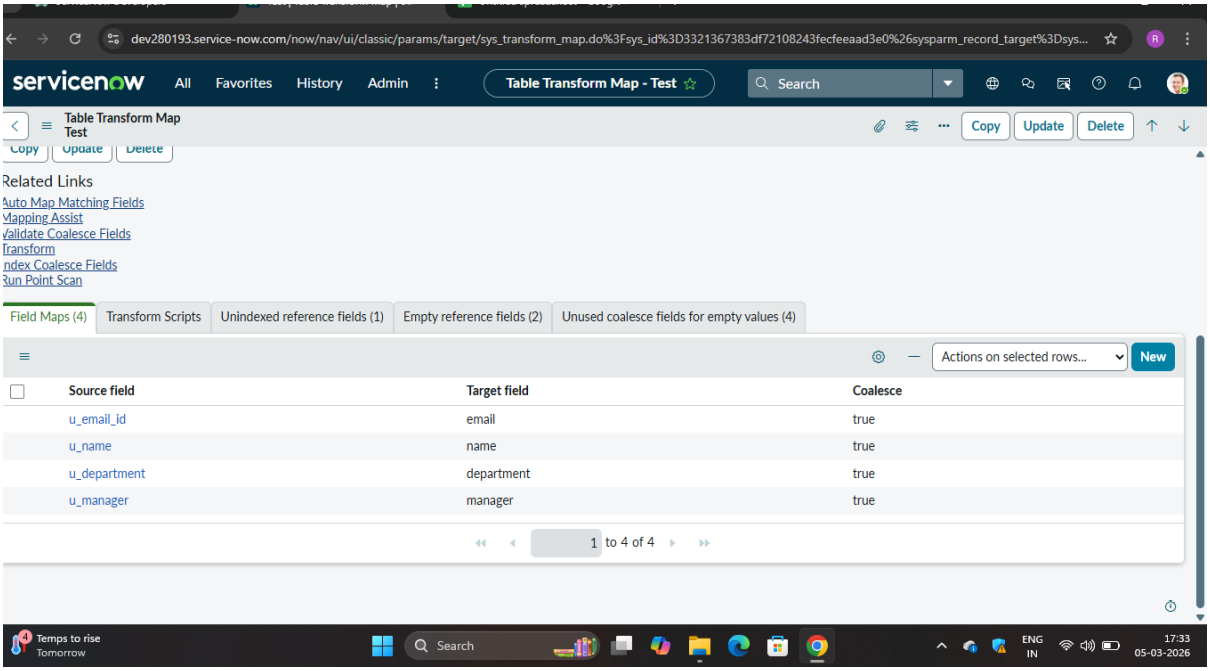
Step 3 – Transform Map Configuration

A Transform Map is created to map fields between the source table and the target table.

Example mapping:

Source Field	Target Field
Name	user_name
Email	email
Department	department

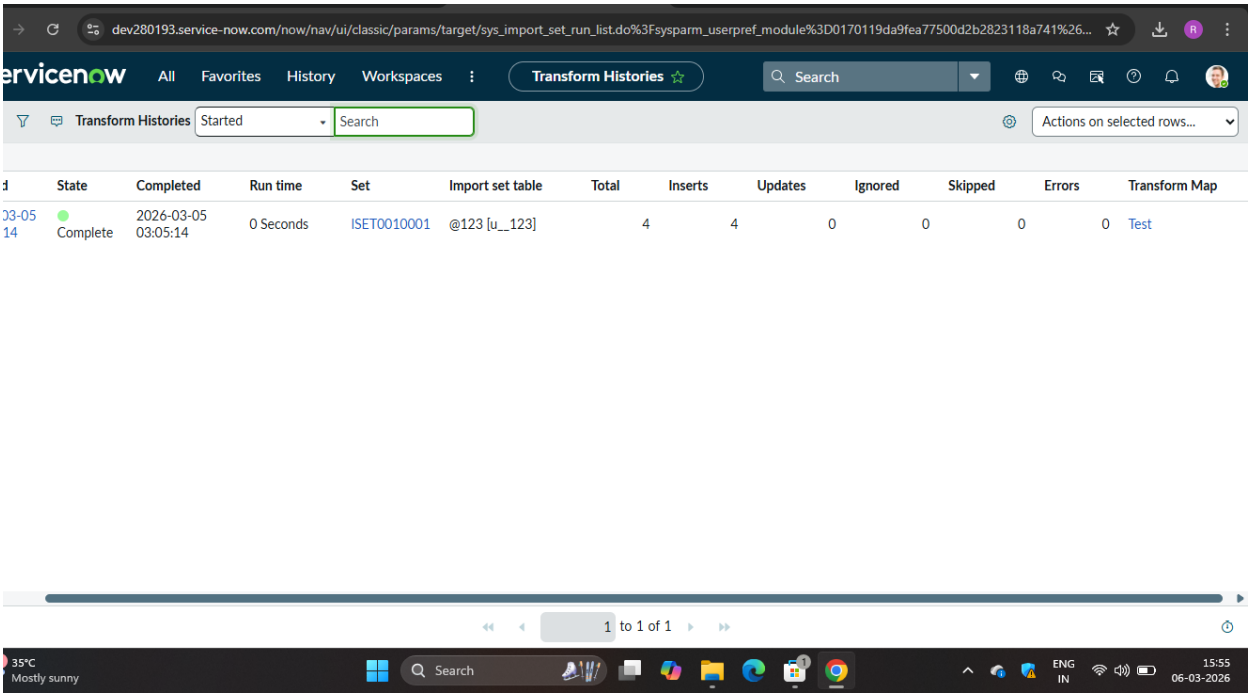
Transform Map moves the data from the Import Set table into the target table.



Step 4 – Data Transformation

After configuring the Transform Map, the transform process is executed.

The system processes the data and inserts the records into the target table.



CHAPTER 5

ACTIVITY DIAGRAM

Start

|

v

Upload Excel File

|

v

Create Import Set

|

v

Load Data into Staging Table

|

v

Create Transform Map

|

v

Map Source Fields

|

v

Run Transform

|

v

Insert Data into Target Table

|

v

Access Data using Dot Walking

|

v

End

Explanation:

1. User uploads an Excel file.

2. Data is stored in Import Set.

3. Transform Map maps fields.

4. Data is moved to the target table.

5. Related data is accessed using Dot Walking.

Explanation:

1. User uploads Excel file.

2. System stores data in Import Set table.

3. Transform Map maps the fields.

4. Data is inserted into the target table.

CHAPTER 6

SEQUENCE DIAGRAM

```
User Target Table      System      Import Set      Transform Map
|                 |                        |                        |
| Upload File |                        |                        |
|-------------->|                        |                        |
|                 | Store Data             |                        |
```

```
 |                  |------------------------->|                          |
 |                  |                          | Apply Mapping            |
 |                  |                          |--------------------------> |
 |                  |                          |                           |
 | Insert Records |                            |                           |
|---------------->|                            |                           |
 |                  |                           |                           |
 |       Data Imported Successfully             |                           |
 ---------------------------------------------------------------------------
```

Explanation:

1. User uploads Excel file.

2. System stores data in Import Set table.

3. Transform Map maps the fields.

4. Data is inserted into the target table.

CHAPTER 7

DOT WALKING CONCEPT

Dot Walking is used to access fields from related tables.

Example:

If an Incident table contains a reference field to the User table, Dot Walking can access user details.

Example syntax:

incident.caller_id.email

incident.caller_id.department

This allows retrieving related information without complex queries.

Advantages:

Easy data access

Reduced query complexity

Better relationship handling

CHAPTER 8

TESTING

Test Case 1

Input: Upload Excel file

Expected Result: Data loaded into Import Set table

Result: Successful

Test Case 2

Input: Execute Transform Map

Expected Result: Data inserted into target table

Result: Successful

Test Case 3

Input: Access reference field

Expected Result: Related data retrieved using Dot Walking

Result: Successful

CHAPTER 9

ADVANTAGES

Advantages of the system:

- Automates data import
- Reduces manual work
- Improves accuracy
- Saves time

- Easy relationship mapping

CHAPTER 10
LIMITATIONS

Some limitations include:

- Requires correct field mapping
- Incorrect source data may cause errors
- Initial setup requires configuration

CHAPTER 11
CONCLUSION

This project demonstrates how data import can be automated using Import Sets and Transform Maps. It also shows how relationships between tables can be accessed using Dot Walking.

The system improves efficiency by reducing manual work and ensuring accurate data processing. Automated data import helps organizations manage large datasets quickly and effectively.